

Segmentation, VPN & ZTNA

Segmentation = blast-radius control

Split network into zones: users / servers / DBs / OT / cloud / guests

Enforce with firewalls per-zone (stateful rules: allow only needed)

Micro-segmentation: app-level segments (a DB only talks to its app server's VPC/NIC)

'Flat network' = every compromise is a total compromise

VPN flavors

VPN	Best for	Notes
IPsec	Site-to-site, legacy	Strong, complex
OpenVPN / WireGuard	Client access	WG: newer, fast, simple keys
SSL VPN (portal/app)	Browser+archive access	Least native, great breadth
Split vs full tunnel	Traffic path choice	split: protect only corp; full: everything througe

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# WireGuard quick (server):
wg genkey | tee privatekey | wg pubkey > publickey
[Interface] Address = 10.0.0.1/24 ListenPort = 51820
[Peer] PublicKey = CLIENT_PUB AllowedIPs = 10.0.0.2/32
```

ZTNA ("VPN replacement")

Scope: *application*, not network. 'App proxy' → your app only.

Identity-based (SSO/MFA) + device posture before session

No network exposure: apps are 'invisible' to anyone else

Cloud-control: Identity provider as the perimeter

Choosing

Small/simple homes: WireGuard or OpenVPN for remote admin

Enterprise: ZTNA for users; site-to-site only for partner links

Legacy apps: SSL-VPN portal; migrate to app-based ZTNA

Craft rules

2FA on every VPN; device compliance check

Least-privilege routes (AllowedIPs per user)

Log sessions; geo/behavior anomaly → challenge

VPNs give a door; ZTNA gives a usher who verifies every visitor and only opens specific rooms. Modern access = usher, not janitor.

← WAF, Reverse Proxies & Conte...

Network Monitoring, NetFlow & ...